

A browser-driven cVEP-based BCI web speller

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Abstract—By utilizing brain signals, the Brain-Computer Interface (BCI) enables non-muscular communication. In recent decades, BCI systems—particularly speller interfaces—have offered a variety of graphical user interfaces (GUIs). Many attempts had been made to improve the system’s user-friendliness and speller speed. In this paper we present a web-based BCI speller based on Code-Modulated Visual Evoked Potentials (cVEPs), which can be accessed at <https://bci-lab.hochschule-rhein-waal.de/en/cvepspeller/>. As a result, this web speller is now available to BCI researchers worldwide free of charge, and can be used with variety of own classifier applications. This web speller was successfully tested with the majority of modern web browsers. In the three-step web speller, each character can be selected by navigating through three distinct web interface screens. In this study the web-based cVEP speller was tested and compared to our “local speller” (incorporating the GUI and the signal processing in one application) by seven subjects, who were asked to spell the words “BCI_LAB” and “KLEVE”. All subjects were able to perform the spelling tasks, with a mean accuracy of 95.02% and an average Information Transfer Rate (ITR) of 42.55 bits/min, compared to our “local speller” with a mean accuracy of 93.81% and an average ITR of 48.58 bits/min, respectively. The results showed similar values, confirming the suitability of the suggested web speller for the representation of the cVEP stimuli.

Index Terms—Brain-Computer Interface (BCI), Web Speller, Code-Modulated Visual Evoked Potential (cVEP), Web Browser

I. INTRODUCTION

Brain-Computer Interfaces (BCIs) provide a new form of communication between the human brain and external devices, by detecting and analyzing brain activity in real time [1], [2]. The development of BCI communication systems has seen significant advancements in recent years, particularly based on visual evoked potentials, e.g. Steady-State Visual Evoked Potentials (SSVEPs) and code-modulated Visual Evoked Potentials (cVEPs). Those BCI paradigms have gained much attention mainly due to their high Information Transfer Rates (ITR) and minimal training requirements [3].

SSVEP spellers decode the commands of the users based on SSVEP responses produced by the users’ brain when gazing on visual stimuli in the form of flickering boxes with different constant frequencies [4]. Code-modulated Visual Evoked Potential (cVEP)-based spellers utilize specific code patterns as stimuli, typically the m-sequences, which are phase-shifted between the individual targets [5].

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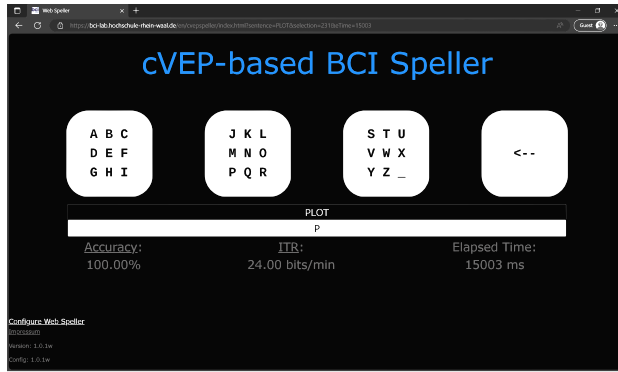
Recent research has focused on designing stimulus sequences tailored to cVEP BCIs, aiming to decrease training time and increase the number of effectively utilized targets [4]. Furthermore, the cVEP speller has demonstrated potential for hands-free communication and has been evaluated for its performance in various applications, including navigation and action selection [6]. Additionally, another study highlights the superior performance of cVEP (40.23 bit/min, 97.83% accuracy) over SSVEP (31.87 bit/min, 95.28% accuracy) [2]. VEP-based BCI spellers are proposed for various application areas, including the medical field for stroke rehabilitation [7], gaming and smart home control [8].

Recent advancements in browser based BCIs have already significantly expanded the accessibility and usability of BCI technologies. BCI2000Web and WebFM have enabled real-time browser-based interactions with human neurophysiological recordings, thereby facilitating web-based BCIs and real-time functional brain mapping [9].

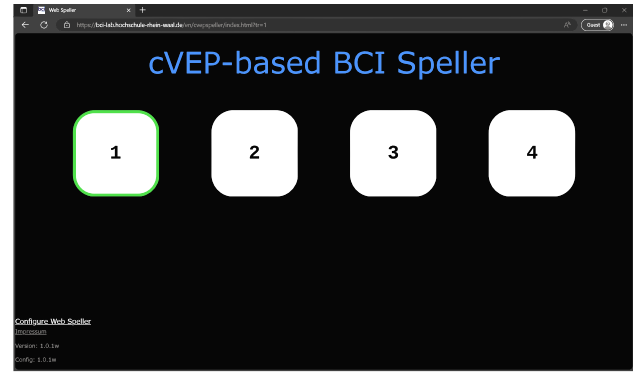
The advancements mentioned have enabled access to BCI applications through web browsers, enhancing practicality and user experience. Moreover, the integration of asynchronous P300-based BCIs with web browsers has enabled individuals with disabilities to access web-based applications, such as word processing and email clients, providing them with enhanced communication and control capabilities [10].

A paper we published in 2018 significantly contributed to this field by introducing a novel approach to SSVEP-based BCI spellers (from then on, an online SSVEP system is available to BCI researchers worldwide free of charge). The paper describes the development of a web-based speller system that underwent successful testing on healthy subjects, yielding encouraging results with average ITR of 12.7 bits/min and the mean accuracy of 94.5% [12]. As in case with the SSVEP online speller, after every command classification the system is retrieving a new URL (along with new query string data) in the web browser.

This paper aims to open up new possibilities and to enhance BCI communication accessibility for everyone with this new browser-driven cVEP-based BCI web speller. The developed cVEP-based web speller interface was tested and compared with the standard offline version (already used many times, and described in details e.g. in [2], “local speller”), with the goal of matching or surpassing the ITR and accuracy values of the latter. To our knowledge this is the first effort to develop and test a cVEP-speller which is driven directly by the web browser. In a study conducted with seven healthy participants it has yielded encouraging results.



(a) Web Speller on “Spelling” mode with letter “P” selected.



(b) Web Speller in “Training” mode, 1st stimulus box highlighted.

Fig. 1: In Fig. 1a, the “Speller” interface shows the letter “P” typed, depicting the sequence of steps during the “selection” query. Fig. 1b showcases the “Training” screen, accessed by entering “tr=1” in the query string. **“Training” screen transforms letters into corresponding box numbers, highlighting the relevant box in green.** Keypress “F” will highlight the next box and keypress “G” will start the flickering for both “Training” and “Speller” screens.

II. METHODS AND MATERIALS

A. Conventional Three-Step-Speller

To establish a benchmark for comparing our web speller, we utilized the cVEP-based Three-Step-Speller previously described in our research [2]. Three-step spellers are highly reliable and robust, requiring only four different visual stimuli. They have been thoroughly tested and proven to work effectively [2]. The graphical user interface (GUI) of this speller is identical to the aforementioned, the interface features four stimuli, equally spaced, to display the sequences, with a text box positioned below them (as shown in the Fig. 1a). **Additionally, it includes a menu bar which allows users to modify various settings such as the words to be typed, gaze shift times, threshold values, and the minimum/maximum time windows.** For a comprehensive understanding of the classification methods employed in this speller, readers are directed to [2].

B. Web Speller’s Uniform Resource Locator (URL)

The Uniform Resource Locator (URL) for the speller interface comprises the host address, the resource path leading to the HTML file, and the associated query string (see Fig. 2).

1) *Host Address*: The host address indicates the name of the host where the resource is stored. It should begin with the protocol (HTTP/HTTPS) to access a resource over the internet.

2) *Resource Path*: The resource path specifies the location of the host’s resource, often identified by its file extension (e.g., “.php”, “.asp”, “.html”, or “.aspx”). If the index page is in the working directory, there’s no need to specify it in the resource path. In our case, we utilized the “index.html” file resource (see Fig. 2).

3) *Query String*: The query string, part of the URL, contains the speller interface data. Notably, the query string’s length for this study will not exceed 256 characters, well below the maximum limit supported by most web browsers (e.g., Microsoft Internet Explorer supports a maximum URL length of 2083 characters).

For the cVEP-based BCI web speller, the query string consists of the following components:

Sentence: The “sentence” section contains the **actual sentence to be typed during copy spelling mode**. This value can be supplied using the configuration stage’s produced cookie or query string.

Selection: The “selection” part includes all **classified commands provided by the classifier**. Each token in the selection is represented by a number from 1 to 4, indicating the classifier’s identification of the stimulus’ index number.

eTime: The “eTime” within the query string indicates the elapsed time since the categorization began, measured in milliseconds, and is among others necessary in order to calculate the information transfer rate (ITR).

tr: The “tr” parameter in the query string serves as a selector for the training mode, with values ranging from 1 to 4, corresponding to discrete boxes in a left-to-right sequence. This input initiates the training mode and determines the initial green-highlighted stimulus box (see Fig. 1b).

C. Three-Step Web Speller

The functionality of cVEP-based three-step web speller is divided into two main parts:

1) *Speller Configuration*: The configuration panel for the web-based speller (accessible at: <https://bci-lab.hochschule-rhein-waal.de/en/cvepspeller/config.html>) was designed to establish fundamental settings for the speller interface. In the configuration panel, users can tailor settings, e.g. stimuli sequences by selecting specific boxes. The two columns of boxes represent circular bit shifts to the left and right direction, respectively. Each row offers a distinct configuration option, specifying the bit shift amount, number of stimuli, and video refresh rate. Users can select only one stimuli option at a time, as each option includes all four associated stimuli.

In the configuration panel, users can input text for the system’s test run. The text must adhere to specific criteria: it must be between 3 and 128 characters in length. Furthermore,

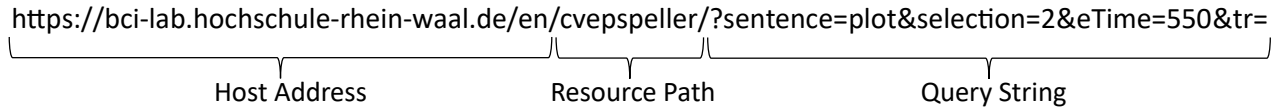


Fig. 2: Parts of the Uniform Resource Locator (URL) for the cVEP-based BCI web speller. Pressing “G” key to initiate.

users are restricted to entering uppercase letters from the English alphabet (A-Z). Spaces in the text are to be substituted with underscores “_”. It is important to note that special characters (% , & , \$, * , ? , @ , # etc.) and numerical digits (0-9) are prohibited. These restrictions are in place due to the constraints imposed by the four-stimuli interface, which necessitates the spelling of words in three steps, and limits the number of available characters. User needs to keep in mind that text input from query string “Sentence” will override the text chosen in the configuration screen and as long as the string is present, the accuracy and ITR measurements will be done according to the word from the query string.

The configuration panel also allows to show or hide the additional information on the speller web page. This includes the information on the accuracy, Information Transfer Rate (ITR) and the elapsed time.

After completing the selection process, users can save the configured settings by clicking the “Save” button, which stores the configuration in HTML cookies. When the user subsequently loads the speller interface from the web server, these settings are retrieved from the persistent cookies stored on the local hard drive of the user’s computer, with no expiration date. To modify the cookie’s value, users need to adjust the settings in the configuration panel and save the cookie once more (it will overwrite the old settings).

2) *Speller Interface*: The main component of the three-step cVEP-based web speller is its speller interface. Now accessible at `https://bci-lab.hochschule-rhein-waal.de/en/cvepspeller/`, the speller web page disseminates content globally, facilitating access for numerous users. Developed with HTML and CSS, the graphical interface embodies a client-side scripting approach, utilizing JavaScript for logical processing within the web browser. This scripting ensures the dynamic functionality of the speller. Parsing query string values becomes pivotal in identifying the classified command index, subsequently updating the speller interface as needed.

Stimuli: The spelling interface incorporates four distinct stimulus options. Four boxes (278 x 278 pixels by default) arranged as 1 x 4 stimulus matrices (see Fig. 1a and Fig. 1b). These flickering cVEP stimuli are presented through WebM video files, which ensure synchronous display by incorporating all four boxes in each frame. Each frame of the video shows the four stimulus boxes, each displaying the corresponding bit from the m-sequence for that frame. The frames are created considering the precise delay required per frame and then combined using FFmpeg to produce the final WebM video, which ensures that the stimuli boxes synchronized with each other and time accurate.

During the design and initial testing phase of the web speller, the stimuli were created using GIF files, similar to our previous SSVEP-based web speller [12]. However, this approach quickly proved to be inaccurate and not fast enough for this application. We were also unable to achieve the 60 Hz frame rate, as the GIF format seems to be limited to 50 Hz. After the unsuccessful attempt with GIF files, we tried to render the stimuli through JavaScript. However, a prominent challenge emerged as the stimuli exhibited jitter (fluctuations around ± 8 ms) during this phase.

These were the main reasons behind the switch to WebM. We have observed some timings issues with WebM videos as well. The initial videos were exactly 1.05 seconds long and sometimes on video loopbacks the video player skipped some frames. In order to minimize this issue, we extended video duration to 10.50 seconds (1.05 seconds x 10). This decision significantly enhanced the accuracy of the stimuli while keeping the file sizes manageable (~293 Kbytes vs. ~30 Kbytes).

The cVEP stimuli used the 63-bit m-sequence [5]

$c_1 = 101011001101110110100100111000$
 $101111001010001100001000001111110,$

whereas “0” denoted “black” and “1” signified “white”, thereby showcasing a state of complete contrast. The remaining stimuli $c_k, k = 2, \dots, K$, where $K = 4$, were generated by circularly left (or right) shifting c_1 by 4, 8 or 16 bits depending on the option selected on the configuration screen. The update rate was set to 60 Hz; thus, the colour of the stimulus changed in accordance with the bit sequence: every first, second or fourth frame depending on the used refresh rate (60, 120, or 240 Hz). Parameters such as bitshift, direction between boxes, and refresh rates are customizable through the configuration screen, allowing for precise control and adjustment in accordance with experimental requirements.

This m-sequence was displayed for a duration of 1.05 s, which emulates to the historical vertical refresh rate of 60 Hz, which was only available on older LCD screens and therefore very popular in many previous studies [11]. In the context of this investigation, we adhere to a circular left shift operation with a 4-bit configuration, implemented at a refresh rate of 240 Hz in order to have a common ground with our local speller.

Training Mode: The activation of the training mode for the web speller can be achieved by assigning a numerical value between 1 and 4 to “tr” query parameter. A numerical scale from 1 to 4 is used to indicate the progressive positioning of the stimuli, with 1 representing the leftmost and 4 representing the rightmost.

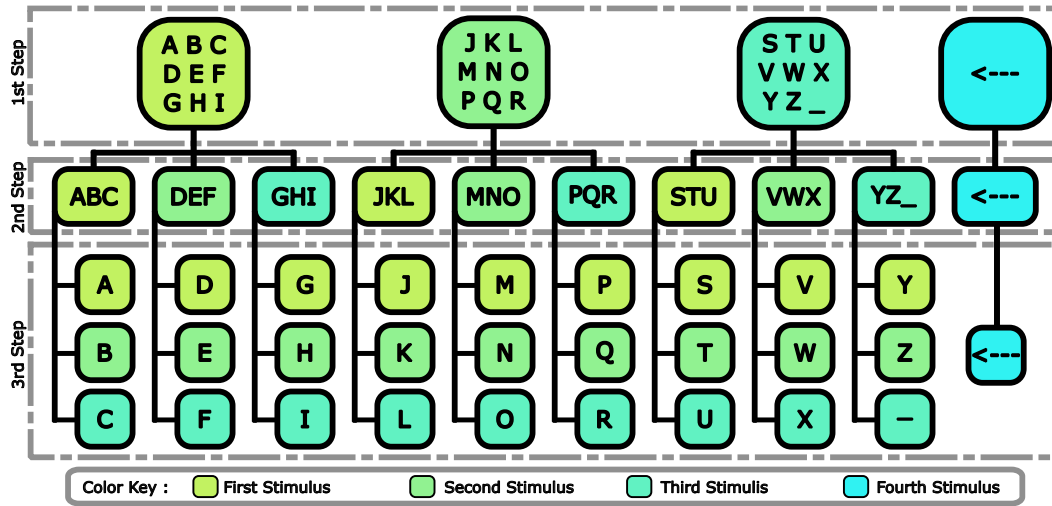


Fig. 3: The “Three-Step Speller Tree” illustrates the selections required to type a letter. The spelling of every individual letter required exactly three steps. In each step, the fourth box represented the undo function which deleted the last spelled character (in step 1) or returned to the previous step (in steps 2 and 3).

Upon page load, four WebM videos with corresponding numbered boxes highlighted in green load simultaneously as it is shown in the Fig. 1b. Only the video matching the query string is initially visible. Pressing the “F” key hides the current video and displays the next video with highlighted box, progressing from the first towards the fourth with every keypress until fourth where it loops back to first video. During the training mode, users can initiate video playback manually and make the stimulus flicker by pressing the “G” key.

Elements such as text boxes, Information Transfer Rate (ITR), accuracy, and “eTime” visuals will be hidden as long as the query parameter “tr” has a value. The stimuli will be chosen according to the cookies that were set by the configuration page. In case of no cookies present, the default stimuli will be used which exhibit a circular leftward bit-shift pattern, where each successive sequence is obtained by circularly left-shifting the preceding sequence by four bits at a video refresh rate of 60 Hz.

Spelling Mode: The web speller will be on spelling mode by default. Keypress “G” should be pressed to start flickering. The use of keypresses minimizes the delay caused by network connections. The characters of the English alphabet, along with underscore (“_”, to represent the space) are overlaid on the first three stimuli. The fourth stimuli is always overlaid with command option of “<—”. The decision tree for the possible overlaid text options is shown in the Fig. 3. According to Fig. 3, all of the first child node values are overlaid on the first stimulus, all of the second child node values are overlaid on the second stimulus, all of the third child node values are overlaid on the third stimulus, the “<—” option text is overlaid on the fourth stimulus. At any stage of the experiment, the text values from the siblings are shown, according to the classified option.

Suppose the user aims to select the character “P”. In the

1st step, the 27 characters comprising the 26 English alphabet letters and the underscore character (“_”) are grouped into three sets, each containing nine characters: {A-I}, {J-R} and {S-Z,_}. These sets are presented in the first three stimuli, while the fourth stimulus displays the option “<—”. With the character “P” now included in the character set of the second stimulus, the user will direct their gaze towards the second stimulus. After the accurate classification, the web speller interface transitions into the 2nd step. In the 2nd step, the text overlaid on the initial three stimuli is modified by subdividing the second set of letters (i.e. {J-R}) into further three sets: {J-L}, {M-O}, {P-R}. As in the 2nd step, the character “P” lies on the third stimulus again, so the user will gaze at the third stimulus. After the correct classification of the command, the speller interface moves to the 3rd step, and the text of the classified third stimulus (PQR) is divided into three single characters “P”, “Q”, “R”, and shown on the first three stimuli. The “<—” option for the fourth stimulus remains the same. Now the user will gaze at the first stimulus to select the character “P”. Upon the accurate classification of the command, the corresponding character is typed on the web speller interface, and the speller reverts to the 1st step (see Fig. 1a). If an incorrect character is typed or a wrong step is taken, the user can use the “<—” option as needed. The web speller interface displays both the text to be spelled and the text that is actually spelled in non-editable text boxes, along with Accuracy (%), ITR (bits/min), and Elapsed Time (ms) during the execution phase of the experiment.

D. Hardware

The used computer was a Dell Precision Desktop with RTX-3070 graphics card that operated on Microsoft Windows 10 (21H2) Education running on an Intel processor i9-10900K. A liquid crystal display screen (Asus ROG Swift PG258Q, Full-HD, 240 Hz maximal refresh rate) was utilised for stimulus presentation.

TABLE I: Results for seven subjects spelling “BCI_LAB” and “KLEVE” using “Web Speller” and “Local Speller”.
Used abbreviations: “Acc.” for accuracy, “ITR” for information transfer rate, and “eTime” for elapsed time.

#	WEB SPELLER						LOCAL SPELLER					
	KLEVE			BCI_LAB			KLEVE			BCI_LAB		
	Acc. [%]	ITR [bits/min]	eTime [s]	Acc. [%]	ITR [bits/min]	eTime [s]	Acc. [%]	ITR [bits/min]	eTime [s]	Acc. [%]	ITR [bits/min]	eTime [s]
1	94.74	55.85	33.05	92.31	42.95	54.00	94.12	46.43	34.80	92.59	46.83	51.95
2	100.00	50.99	35.30	95.65	45.67	50.55	94.12	60.40	26.75	89.66	48.96	48.20
3	100.00	51.95	34.65	95.65	45.67	50.55	100.00	70.45	25.55	89.66	46.73	50.50
4	100.00	52.17	34.50	100.00	43.95	57.34	100.00	54.30	33.15	95.65	47.12	49.00
5	100.00	59.50	30.25	100.00	44.15	57.08	100.00	68.31	26.35	100.00	65.63	38.40
6	86.96	17.90	95.20	82.86	20.02	111.95	90.48	38.22	46.00	87.10	36.06	64.00
7	89.47	31.23	49.20	92.59	33.69	72.20	84.00	15.75	105.90	96.00	35.03	72.55
Avg.	95.88	45.66	44.59	94.15	39.45	64.81	94.67	50.55	42.65	92.95	46.62	53.51
S.D.	5.63	15.19	23.12	5.86	9.51	22.05	6.02	19.16	28.81	4.52	10.11	11.26

An EEG amplifier (g.USBamp, Guger Technologies, Graz, Austria) was used, utilizing all 16 signal channels which were placed according to the international system of EEG electrode placement: P₇, P₃, P_Z, P₄, P₈, PO₇, PO₃, PO_Z, PO₄, PO₈, O₁, O_Z, O₂, O₉, I_Z, and O₁₀. Additionally, the reference electrode was positioned at C_Z, while the ground electrode was placed at AF_Z. During the preparation stage, an abrasive electrolytic electrode gel was used between the electrodes and the scalp to reduce impedances to below 5 kΩ.

E. Signal Classification

For the cVEP system, spatial filters were created with the data collected during the recording stage. In this respect, canonical correlation analysis (CCA) was applied to the training trials. In general, CCA can be applied to two multi-dimensional variables $\mathbf{X} \in \mathbb{R}^{p \times s}$ and $\mathbf{Y} \in \mathbb{R}^{q \times s}$ in order to analyze their relationship. Further details about used cVEP signal processing methods used on our local speller can be found in [2].

The signal classification functionality, based on the aforementioned methods, was implemented in Visual C++ for the “local speller” [2]. Upon classification, the index of the classified frequency was added to the “sentence” parameter of the query string, and the modified URL was subsequently submitted to the web browser.

F. Subjects

Seven participants (one female) were recruited for this study; the average age of the participants was 23.36 years, with a standard deviation (SD) of ± 2.4 . All participants provided written consent in adherence to the Declaration of Helsinki, and the study received approval from the ethical committee of the medical faculty at the University Duisburg-Essen. The collected data for analysis purposes was stored anonymously, ensuring the confidentiality of the participants. Notably, four participants had prior familiarity with cVEP-based BCI and had previously tested BCI systems.

G. Experimental Protocol

The online experiment was conducted in the BCI-Lab of Rhine-Waal University of Applied Sciences (HSRW). Participants were briefed about the procedure and the operation of the three-step spellers. Threshold values were adjusted if needed. Once the thresholds were set, participants conducted a test run of the system to familiarize themselves with the spellers. After the familiarization phase, the subjects performed the actual test of the system, and the results were recorded. Initially, participants trained on our local speller’s training screen. Subsequently, they spelled the words “BCI_LAB” and “KLEVE” on our local software’s speller screen, with the starting word determined by subject number. Following the local spelling, the local speller’s screens were replaced with web spellers displayed on Microsoft Edge. After completing the web speller training, subjects repeated the spelling of “BCI_LAB” and “KLEVE” based on their subject number, with our local software running in the background for classifications. Spelling phases concluded automatically upon correct word spelling. On average, each subject’s spelling session (just spelling) lasted 15 to 20 minutes.

III. RESULTS

The BCI system’s performance was evaluated using accuracy (Acc.) and Information Transfer Rate (ITR).

1) *Accuracy*: The accuracy was calculated by dividing the total number of correct commands, including user necessary corrections during speller execution, by the overall commands classified. The accuracy was displayed in percent.

2) *ITR*: The Information Transfer Rate (ITR) was calculated in bits per minute (bits/min) using the given formula [1]:

$$B = \log_2 N + P \log_2 P + (1 - P) \log_2 \left[\frac{1 - P}{N - 1} \right], \quad (1)$$

where:

B = information transferred in bits,

N = number of targets (for this study it is equal to 4),

P = classification accuracy.

To obtain the ITR in bits/min, B is multiplied by the average classification time in minutes. Table I presents the summarized results for all subjects. The table lists the accuracy, the ITR and the time taken by the users to complete the spelling tasks “BCI_LAB” and “KLEVE” using our web speller and local speller.

For the web speller, one of the participants achieved 100% accuracy with the final query string “?sentence=KLEVE&selection=212213122321122&eTime=30250” resulting in an impressive ITR value of 59.50 bits/min.

In contrast, another participant achieved 86.96% accuracy with the query string “?sentence=KLEVE&selection=21221311421424122321122&eTime=95200” yielding an ITR of 17.90 bits/min. Both participants were attempting to type “KLEVE” and a closer examination of the selection parameters reveals the made errors.

IV. DISCUSSION AND CONCLUSION

The primary objective of the study was to introduce a browser-driven three-step cVEP-based BCI speller and compare its performance with our local software already successfully tested multiple times. The proposed speller underwent testing with seven subjects, and all subjects successfully completed spelling tasks, resulting in a 100% literacy rate.

The web speller showed excellent typing accuracy, averaging 95.02%, a 1.20% improvement over local speller. Most participants achieved 100% accuracy, resulting in a mean ITR of 39.45 bits/min for “BCI_LAB” and 45.66 bits/min for “KLEVE” with the web speller, compared to 46.62 bits/min and 50.55 bits/min for the local speller. Our local speller had a 6.03 bits/min higher average ITR, but the web speller still demonstrated significantly high ITR values, representing a substantial improvement over our previous SSVEP based web speller with an average ITR of 12.7 bits/min [12].

In the early iterations of our system, we confronted an issue stemming from fixed delays employed to initiate stimulus flickering. These delays, initially adjusted according to gaze shift values and training times, resulted in persistent network-related timing issues, consistently yielding suboptimal results. In response, we opted to eliminate the delays altogether, implementing key press algorithms to facilitate the toggling of flickering and the alteration of the highlighted stimulus.

In this study, the classifier program retrieves a new URL, along with updated query string data, with each command classification, providing flexibility by altering query string parameter values. In future studies, this approach could be substituted with a publish/subscribe system or PUSH technology utilizing the Comet Application Model.

In conclusion, while the cVEP-based BCI web speller shows a slightly lower average ITR compared to the local version, it remains a viable typing method, especially when compared to our previous SSVEP-based web speller. It can be seamlessly integrated into web-based email and social media interfaces. Future tests with patients are essential to validate its functionality for individuals with disabilities.

Additionally, the web speller allows researchers to perform spelling tasks online, independent of operating systems, classifier applications, and EEG amplifier devices, thus opening numerous possibilities for further research.

We encourage you to try the online web speller by visiting the following link:

<https://bci-lab.hochschule-rhein-waal.de/en/cvepspeller/>

The web speller will be on spelling mode by default. Keypress “G” should be pressed to start flickering.

REFERENCES

- [1] J. R. Wolpaw, N. Birbaumer, D. J. McFarland, G. Pfurtscheller, and T. M. Vaughan, “Brain-computer interfaces for communication and control,” *Clinical Neurophysiology*, vol. 113, no. 6, pp. 767–791, Jun. 2002. [Online]. Available: [http://dx.doi.org/10.1016/S1388-2457\(02\)00057-3](http://dx.doi.org/10.1016/S1388-2457(02)00057-3)
- [2] I. Volosyak, A. Rezeika, M. Benda, F. Gembler, and P. Stawicki, “Towards solving of the Illiteracy phenomenon for VEP-based brain-computer interfaces,” *Biomedical Physics & Engineering Express*, vol. 6, no. 3, pp. 035034, Apr. 2020. [Online]. Available: <https://doi.org/10.1088/2057-1976/ab87e6>
- [3] X. Chen, Y. Wang, S. Gao, T.-P. Jung, and X. Gao, “Filter bank canonical correlation analysis for implementing a high-speed SSVEP-based brain computer interface,” *Journal of Neural Engineering*, vol. 12, no. 4, pp. 046008, Jun. 2015. [Online]. Available: <https://doi.org/10.1088/1741-2560/12/4/046008>
- [4] A. Rezeika, M. Benda, P. Stawicki, F. Gembler, A. Saboor, and I. Volosyak, “Brain-Computer Interface Spellers: A Review,” *Brain Sciences*, vol. 8, no. 4, pp. 57, Mar. 2018. [Online]. Available: <https://doi.org/10.3390/brainsci8040057>
- [5] P. Stawicki and I. Volosyak, “cVEP Training Data Validation-Towards Optimal Training Set Composition from Multi-Day Data,” *Brain Sciences*, vol. 12, no. 2, pp. 234, Feb. 2022. [Online]. Available: <https://doi.org/10.3390/brainsci12020234>
- [6] F. W. Gembler, M. Benda, A. Rezeika, P. R. Stawicki, and I. Volosyak, “Asynchronous c-VEP communication tools-efficiency comparison of low-target, multi-target and dictionary-assisted BCI spellers,” *Scientific Reports*, vol. 10, no. 1, Oct. 2020. [Online]. Available: <https://doi.org/10.1038/s41598-020-74143-4>
- [7] Y. Miao, S. Chen, X. Zhang, J. Jin, R. Xu, I. Daly, J. Jia, X. Wang, and A. Cichocki, “BCI-Based Rehabilitation on the Stroke in Sequela Stage,” *Neural Plasticity*, vol. 2020, pp. 1–10, Dec. 2020. [Online]. Available: <http://dx.doi.org/10.1155/2020/8882764>
- [8] A. Saboor, A. Rezeika, P. Stawicki, F. Gembler, M. Benda, T. Grunenber, and I. Volosyak, “SSVEP-Based BCI in a Smart Home Scenario,” in *Advances in Computational Intelligence*. Springer International Publishing, 2017, pp. 474–485. [Online]. Available: https://doi.org/10.1007/978-3-319-59147-6_41
- [9] G. Milsap, M. Collard, C. Coogan, and N. E. Crone, “BCI2000Web and WebFM: Browser-Based Tools for Brain Computer Interfaces and Functional Brain Mapping,” *Frontiers in Neuroscience*, vol. 12, Feb. 2019. [Online]. Available: <http://dx.doi.org/10.3389/fnins.2018.01030>
- [10] V. Martínez-Cagigal, J. Gómez-Pilar, D. Alvarez, and R. Hornero, “An Asynchronous P300-Based Brain-Computer Interface Web Browser for Severely Disabled People,” *IEEE Transactions on Neural Systems and Rehabilitation Engineering*, vol. 25, no. 8, pp. 1332–1342, 2017. [Online]. Available: <https://doi.org/10.1109/TNSRE.2016.2623381>
- [11] L. Henke, P. Rulffs, F. Adepoju, P. Stawicki, A. Cantürk, and I. Volosyak, “Investigating the Influence of Background Music on the Performance of a cVEP-based BCI,” in *2023 IEEE International Conference on Systems, Man and Cybernetics (SMC)*, Oct. 2023, pp. in press. [Online]. Available: <https://doi.org/10.1109/SMC.2023.9567895>
- [12] A. Saboor, F. Gembler, M. Benda, P. Stawicki, A. Rezeika, R. Grichnik, and I. Volosyak, “A Browser-Driven SSVEP-Based BCI Web Speller,” in *2018 IEEE International Conference on Systems, Man, and Cybernetics (SMC)*, Oct. 2018. [Online]. Available: <https://doi.org/10.1109/smc.2018.00115>